# Practice Quiz 11C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 11

*Adding zero does not change an infinite cardinality.*

1. Let $\mathbf{N}_0 = \{0, 1, 2, \ldots\}$ and $\mathbf{Z}^+ = \{1, 2, 3, \ldots\}$. Define a bijection $h : \mathbf{Z}^+ \to \mathbf{N}_0$ and verify that it is one-to-one and onto.

   **Solution** Set $h(n) = n - 1$. Since $n \geq 1$, $h(n)$ is a nonnegative integer, so the rule maps into $\mathbf{N}_0$. If $h(m) = h(n)$, then $m - 1 = n - 1$, hence $m = n$; it is one-to-one. For every $a \in \mathbf{N}_0$, the positive integer $n = a + 1$ satisfies $h(n) = a$; it is onto. Thus $h$ is a bijection and $\mathbf{Z}^+ \sim \mathbf{N}_0$. ∎

*Solutions prepared using GPT-6 in the Codex harness.*